

Power BI Assignment

Question1:

Power BI is a business intelligence and data visualization tool developed by Microsoft. It helps users connect, analyze, visualize, and share data through interactive dashboards and reports.

Power BI is used in businesses:

Data Visualization – Converts raw data into charts, graphs, and dashboards.

1. Better Decision Making – Helps management make data-driven decisions.
2. Real-Time Insights – Provides live updates and monitoring.
3. Data Integration – Connects with Excel, SQL, cloud services, and many other sources.
4. Easy Report Sharing – Reports can be shared across teams and departments.
5. Automation – Reduces manual reporting work.

Question 2:

1. Power BI Desktop

A Windows application used to:

- Import and transform data
- Create data models
- Design reports and dashboards

It is mainly used by report developers and analysts.

2. Power BI Service

A cloud-based online service where reports are:

- Published
- Shared
- Collaborated on
- Accessed from anywhere

Used for teamwork and report distribution.

3. Power BI Mobile

A mobile application available for Android and iOS devices that allows users to:

- View reports
- Monitor dashboards
- Access business insights anytime

Question 3:

The Power BI workflow follows these steps:

Step 1: Connect Data

Import data from sources such as:

- Excel
- CSV files
- SQL databases
- Web sources
- Cloud platforms

Step 2: Transform Data

Use Power Query Editor to:

- Clean data
- Remove errors
- Handle missing values
- Format columns

Step 3: Create Data Model

Build relationships between tables and organize the data structure.

Step 4: Create Visualizations

Design:

- Charts
- Graphs
- Maps
- KPIs
- Dashboard

Step 5: Generate Reports

Combine visuals into interactive reports.

Step 6: Publish Reports

Upload reports to Power BI Service.

Step 7: Share and Collaborate

Share dashboards with teams and stakeholders for business insights.

Question 4:

Power Query allows several data cleaning operations such as:

1. Remove Duplicate Rows
Eliminates repeated records from the dataset.
2. Handle Missing Values
Replace null values or remove empty rows.
3. Change Data Types
Convert columns into proper formats such as text, number, or date.
4. Split Columns
Split one column into multiple columns using delimiters.

Other tasks include filtering rows, merging tables, trimming spaces, and renaming columns.

Question 5:

Steps to Load Dataset into Power BI

Step 1:

Download the dataset from the provided Google Drive link.

Step 2:

Open Power BI Desktop.

Step 3:

Click on Home → Get Data.

Step 4:

Select the appropriate file type:

- Excel Workbook (if **.xlsx**)
- Text/CSV (if **.csv**)

Step 5:

Browse and select the downloaded dataset file.

Step 6:

Click Open.

Step 7:

A preview window will appear showing the dataset.

Step 8:

Click:

- Load → directly import data, OR
- Transform Data → clean and modify data before loading.

Step 9:

The dataset will now appear in the Fields pane of Power BI.

Question 6:

1. Data View

Data View displays the dataset in table format.

Purpose:

- Inspect rows and columns
- Verify data
- Create calculated columns

2. Report View

Report View is used to create visualizations and dashboards.

Purpose:

- Design reports
- Add charts and graphs
- Create interactive dashboards

3. Model View

Model View shows relationships between tables.

Purpose:

- Manage table relationships
- Build data models
- Organize schema structure

Question 7:

Power BI supports many types of data sources, including:

1. File Sources

- Excel
- CSV
- XML
- PDF
- JSON

2. Database Sources

- SQL Server
- MySQL
- Oracle
- PostgreSQL

3. Cloud Services

- Google Analytics
- Azure
- SharePoint
- Salesforce

4. Online Services

- Web APIs
- Online websites
- OData feeds

5. Big Data Sources

- Hadoop
- Spark

This flexibility allows businesses to combine data from multiple platforms into one dashboard.

Question 8:

Step 1:

Open Power Query Editor by clicking:
Home → Transform Data

Step 2:

Select the Owner Name column.

Step 3:

Go to:
Transform → Split Column → By Delimiter

Step 4:

Choose:

- Delimiter: Space
- Split at: Left-most delimiter

Step 5:

Click OK

Step 6:

Rename the new columns:

- First Name
- Last Name

Untitled - Power Query Editor

File Home Transform Add Column View Tools Help

Close & Apply New Source Recent Enter Data Data source settings Manage Parameters Export query results Refresh Preview Properties Advanced Editor Choose Remove Columns Keep Remove Rows Reduce Rows Sort Split Column Group By Data Type: Text Use First Row as Headers Replace Values Merge Queries Append Queries Combine Files

fx = Table.RenameColumns(#"Changed Type1",{{"Owner_Name.1", "First_Name"}, {"Owner_Name.2", "Last_Name"}})

	Area_sqft	Bedrooms	Bathrooms	First_Name	Last_Name	Contact
1	850	2		Rahul	Mehta	9876543210
2	720	2		Anita	Sharma	9898989898
3	900	3		Ramesh	Kumar	9123456789
4	880	3		Neha	Verma	9988776655
5	950	3		Amit	Joshi	9090909090
6	800	2				9345678901
7	null	2		Suresh	Reddy	9012345678
8	870	3		Divya	Iyer	9556677889
9	1200	4		Manoj	Patil	9887766554
10	700	2		Raj	Singh	null
11	850	2		Rahul	Mehta	9876543210
12	720	2		Anita	Sharma	9898989898
13	900	3		Ramesh	Kumar	9123456789
14	880	3		Neha	Verma	9988776655
15	950	3		Amit	Joshi	9090909090
16	800	2				9345678901
17	null	2		Suresh	Reddy	9012345678
18	870	3		Divya	Iyer	9556677889
19	1200	4		Manoj	Patil	9887766554
20	700	2		Raj	Singh	null
21	850	2		Rahul	Mehta	9876543210

Query Settings

PROPERTIES

Name

Real_Estate_Data_100_Rows

APPLIED STEPS

Source

Promoted Headers

Changed Type

Split Column by Delimiter

Changed Type1

Renamed Columns

Untitled - Power Query Editor

File Home Transform Add Column View Tools Help

Close & Apply New Source Recent Enter Data Data source settings Manage Parameters Export query results Refresh Preview Properties Advanced Editor Choose Remove Columns Keep Remove Rows Reduce Rows Sort Split Column Group By Data Type: Whole Number Use First Row as Headers Replace Values Merge Queries Append Queries Combine Files

fx = Table.TransformColumnTypes(#"Promoted Headers",{{"Property_ID", Int64.Type}, {"Property_Name", type text}, {"City", type text}})

	Area_sqft	Bedrooms	Bathrooms	Owner_Name	Contact
1	12500000	850	2	Rahul Mehta	9876543210
2	9800000	720	2	Anita Sharma	9898989898
3	8500000	900	3	Ramesh Kumar	9123456789
4	11000000	880	3	Neha Verma	9988776655
5	null	950	3	Amit Joshi	9090909090
6	7200000	800	2		9345678901
7	6700000	null	2	Suresh Reddy	9012345678
8	7900000	870	3	Divya Iyer	9556677889
9	13500000	1200	4	Manoj Patil	9887766554
10	6500000	700	2	Raj Singh	null
11	12500000	850	2	Rahul Mehta	9876543210
12	9800000	720	2	Anita Sharma	9898989898
13	8500000	900	3	Ramesh Kumar	9123456789
14	11000000	880	3	Neha Verma	9988776655
15	null	950	3	Amit Joshi	9090909090
16	7200000	800	2		9345678901
17	6700000	null	2	Suresh Reddy	9012345678
18	7900000	870	3	Divya Iyer	9556677889

Query Settings

PROPERTIES

Name

Real_Estate_Data_100_Rows

APPLIED STEPS

Source

Promoted Headers

Changed Type

10 COLUMNS, 100 ROWS Column profiling based on top 1000 rows

PREVIEW DOWNLOADED AT 6:31 PM

The Owner Name column was successfully split into First Name and Last Name using Power Query in Power BI.

